lab10

Max Wang

Table of contents

PDB Statistics 1
Using Molstar 2
Bio3D in R 6
Normal Mode Analysis 8
Comparative structure analysis of Adenylate Kinase 9

PDB Statistics

The Proteinn Data Bank (PDB) is the main repository of biomolecular structural data.

```r
data <- read.csv("pdb_stats.csv", row.names = 1)
head(data)
```

```
                        X.ray    EM   NMR Integrative Multiple.methods Neutron
Protein (only)         178795 21825 12773         343               226      84
Protein/Oligosaccharide  10363  3564    34           8                11       1
Protein/NA                9106  6335   287          24                 7       0
Nucleic acid (only)       3132   221  1566           3                15       3
Other                      175    25    33           4                 0       0
Oligosaccharide (only)      11     0     6           0                 1       0
                       Other  Total
Protein (only)            32 214078
Protein/Oligosaccharide    0  13981
Protein/NA                 0  15759
Nucleic acid (only)        1   4941
Other                      0    237
Oligosaccharide (only)     4     22
```

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
c.sums <- colSums(data)
c.sums/c.sums["Total"] *100
```

```
           X.ray               EM              NMR      Integrative
     80.95077464      12.83842935       5.90278614       0.15340257
Multiple.methods          Neutron            Other            Total
      0.10441012       0.03533881       0.01485836     100.00000000
```

80.95% of the PDB structures are solved by X-ray and 12.84% are solved by electron microscopy.

> Q2: What proportion of structures in the PDB are protein?

```
sum(data$Total[1:3])/c.sums["Total"] *100
```

```
  Total
97.9118
```

97.91% of all structures in the PDB database are proteins.

> Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1,173 structures for HIV proteases.

Using Molstar

Using the molstar viewer we can look to view proteins of interest and highlight components.

With Mol* visual representations can be made like:

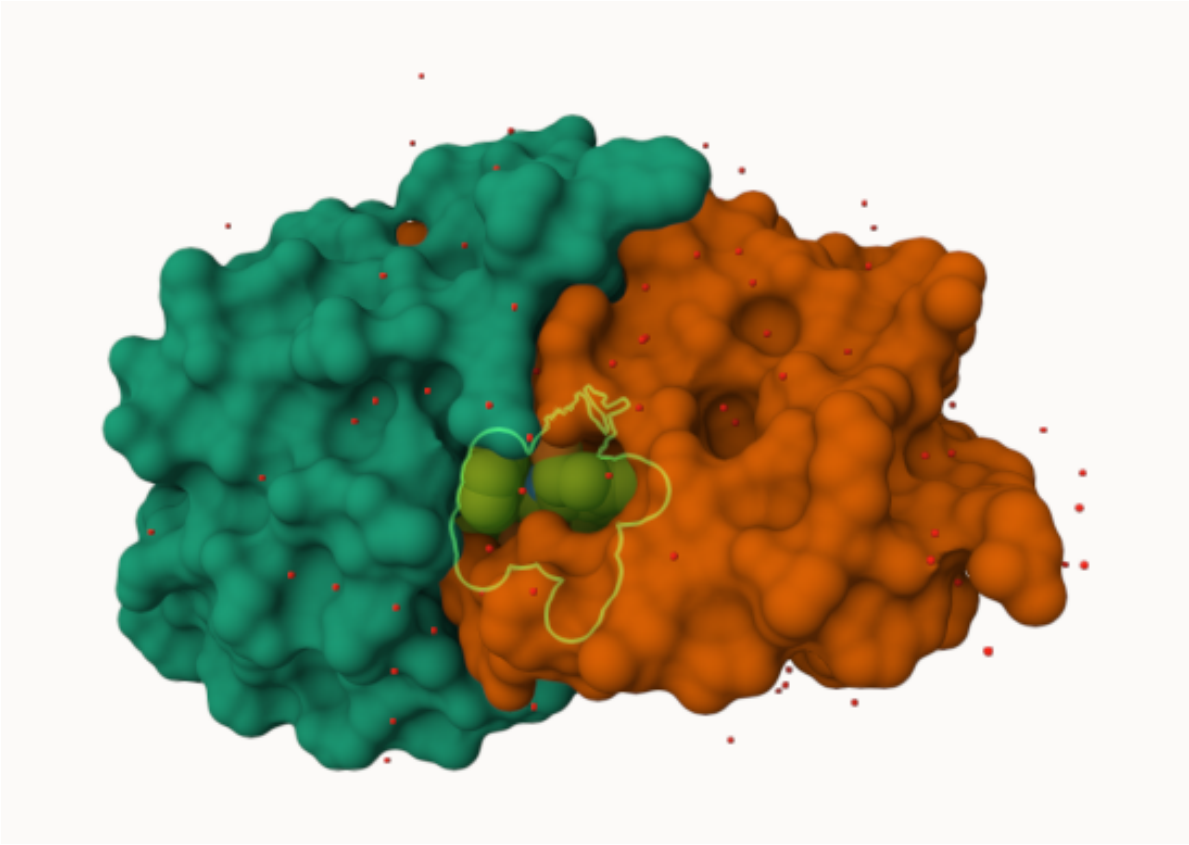


Figure 1: first view of HIV-Pr

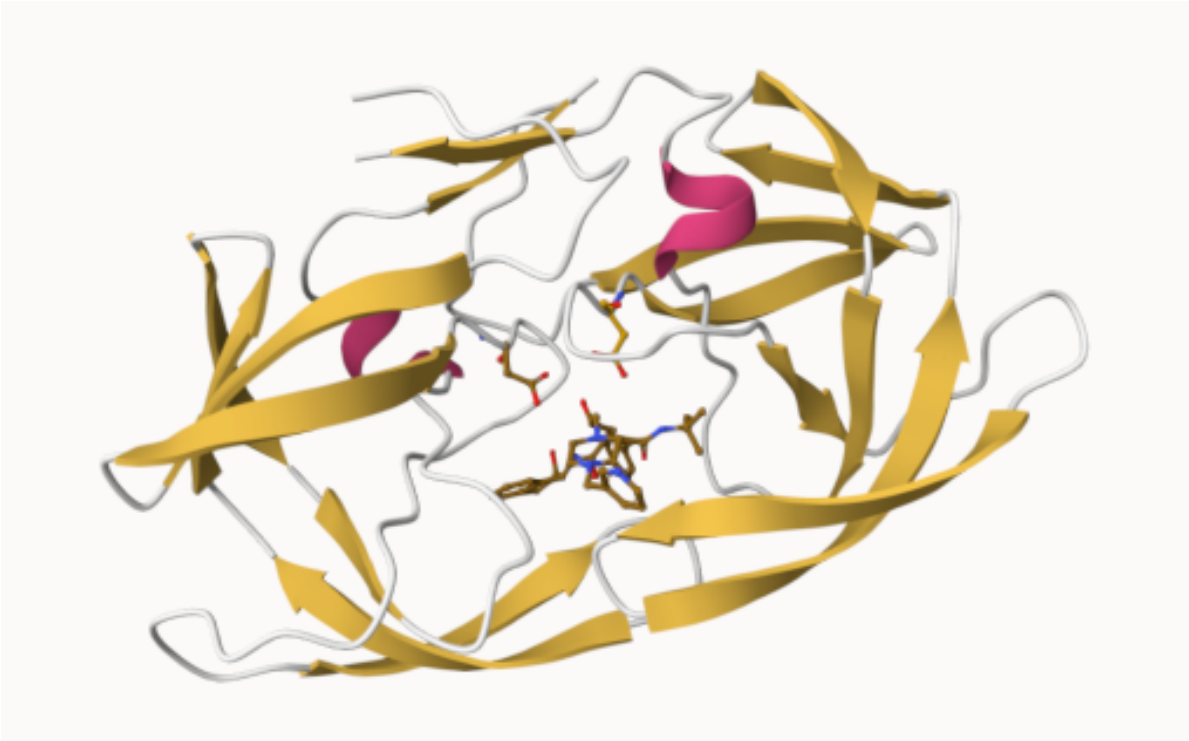


Figure 2: Ribbon representation colored by secondary strucutre

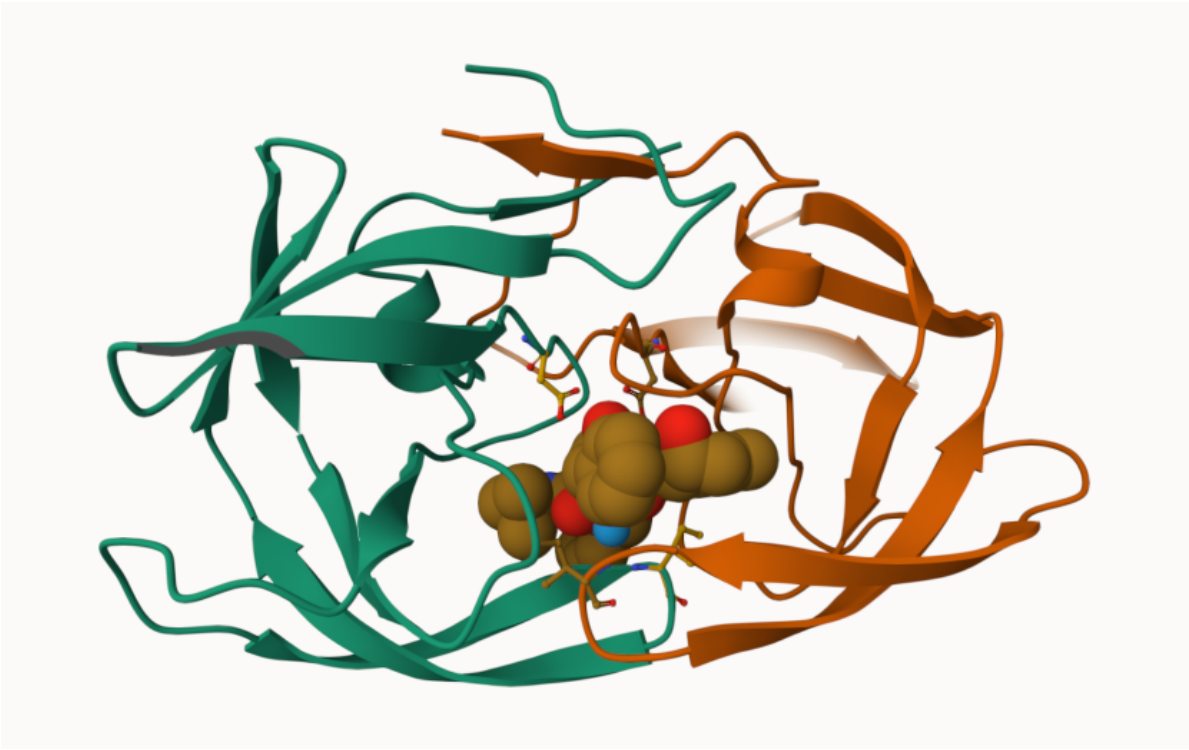
> Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Only 1 atom is visible because otherwise water would cause too much visual clutter.

> Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The water molecule has interactions with residue number ile50 because it forms hydrogen bonds with the ile50 backbone on both subunits.

> Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.



Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

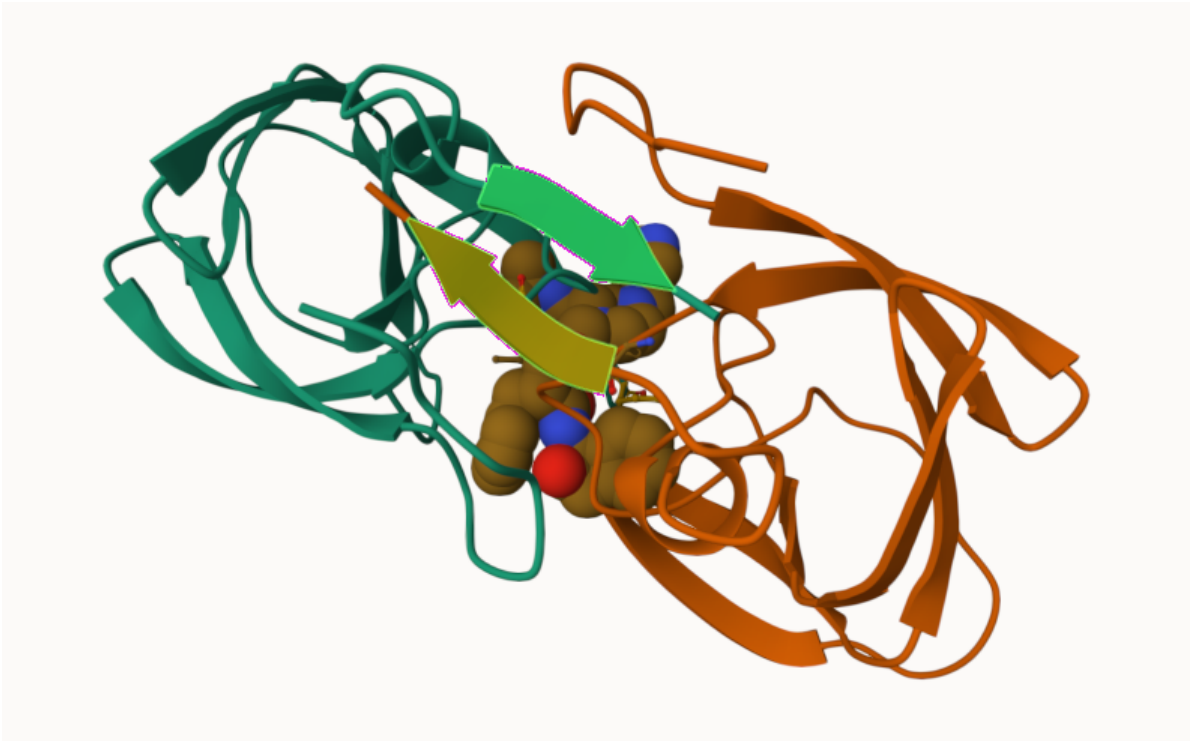


Figure 3: The beta sheet at the top of the protease can only form when both monomers combine since it uses protions from both of the subunits.

Bio3D in R

Accessing pdb data and loading the bio3d package

```
library(bio3d)
```

```
Warning: package 'bio3d' was built under R version 4.4.3
```

```
hiv <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

HOH (water)

> Q9: How many protein chains are in this structure?

2

Looking into the atom records

```
head(hiv$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

Exploring the `bio3dview` package that isnt on CRAN. Using the `remotes` package to install a package from GitHub.

```
library(bio3dview)
library(NGLVieweR)
```

```
Warning: package 'NGLVieweR' was built under R version 4.4.3
```

```
#view.pdb(hiv) |>
  #setSpin()
```

Highlighting certain residues

```
#sele <- atom.select(hiv, resno=25)
#view.pdb(hiv, cols=c("navy","teal"),
         #highlight = sele,
         #highlight.style = "spacefill") |>
  #setRock()
```

Normal Mode Analysis

Using normal mode analysis to predict the dynamics of the protein Adenylate Kinase

```
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

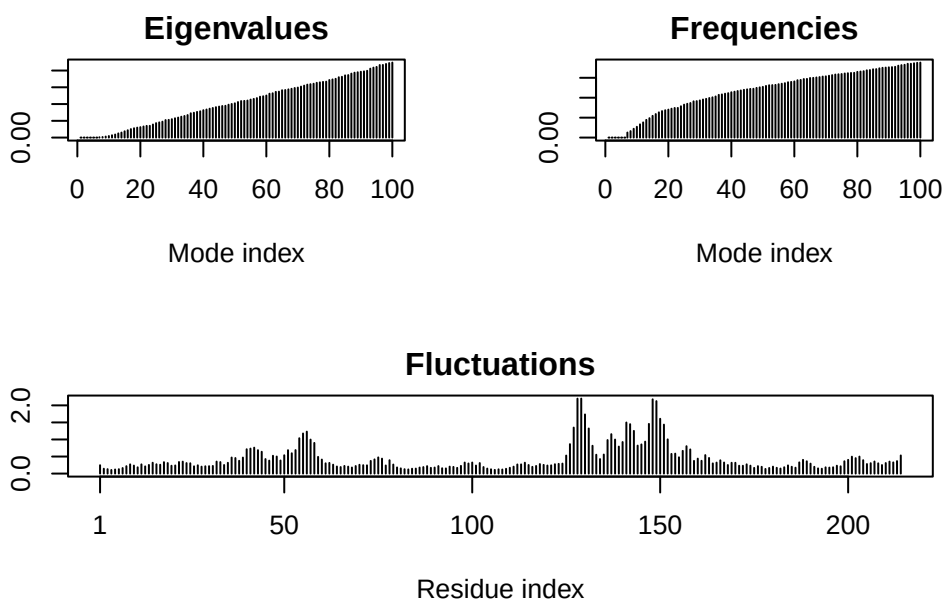
using the normal mode analyiss function on the adk object.

```
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.03 seconds.
 Diagonalizing Hessian...   Done in 0.36 seconds.
```

Plotting results

```
plot(m)
```



Storing the data as a movie to be viewed in Mol*

```r
mktrj(m, file="adk_m7.pdb")
```

Comparative structure analysis of Adenylate Kinase

Returning to HTML outputs.

accessing the pdb data for adenylate kinase

```r
aa <- get.seq("1ake_A")
```

```
Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.
```

```r
aa
```

```
              1        .         .         .         .         .         60
pdb|1AKE|A    MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
              1        .         .         .         .         .         60

             61        .         .         .         .         .         120
pdb|1AKE|A    DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
             61        .         .         .         .         .         120

            121        .         .         .         .         .         180
pdb|1AKE|A    VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
            121        .         .         .         .         .         180

            181        .         .         .   214
pdb|1AKE|A    YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
            181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

Feeding aa sequence into blast

```
blast <- blast.pdb(aa)
```

```
 Searching ... please wait (updates every 5 seconds) RID = STMYZ2WA014
 ....
 Reporting 96 hits
```

```
head(blast$hit.tbl)
```

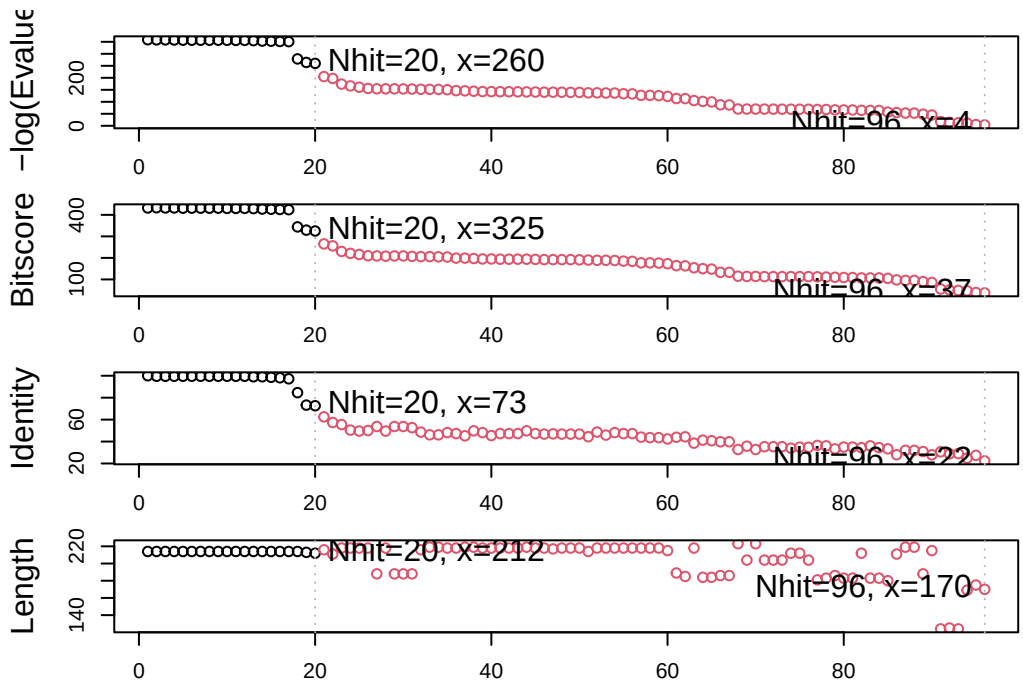
```
        queryid subjectids identity alignmentlength mismatches gapopens q.start
1 Query_4995653     1AKE_A  100.000             214          0        0       1
2 Query_4995653     8BQF_A   99.533             214          1        0       1
3 Query_4995653     4X8M_A   99.533             214          1        0       1
4 Query_4995653     6S36_A   99.533             214          1        0       1
5 Query_4995653     9R6U_A   99.533             214          1        0       1
6 Query_4995653     9R71_A   99.533             214          1        0       1
  q.end s.start s.end    evalue bitscore positives mlog.evalue pdb.id    acc
1   214       1   214 1.78e-156      432    100.00    358.6267 1AKE_A 1AKE_A
2   214      21   234 2.91e-156      433    100.00    358.1351 8BQF_A 8BQF_A
3   214       1   214 3.18e-156      432    100.00    358.0464 4X8M_A 4X8M_A
4   214       1   214 4.67e-156      432    100.00    357.6621 6S36_A 6S36_A
5   214       1   214 1.04e-155      431     99.53    356.8615 9R6U_A 9R6U_A
6   214       1   214 1.23e-155      431     99.53    356.6937 9R71_A 9R71_A
```

plotting the output into variable hits

```
hits <- plot(blast)
```

```
  * Possible cutoff values:    260 3
            Yielding Nhits:    20 96

  * Chosen cutoff value of:    260
            Yielding Nhits:    20
```



Top ids

```
head(hits$pdb.id)
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A"
```

Downloading the "top hits" which will be ADK strucutres from the pdb database.

```
files <- get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/1AKE.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/8BQF.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/4X8M.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/6S36.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/9R6U.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/9R71.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/8Q2B.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/8RJ9.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/6RZE.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/4X8H.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/3HPR.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/1E4V.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/5EJE.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/1E4Y.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/3X2S.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/6HAP.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/6HAM.pdb exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/8PVW.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/4K46.pdb exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = T, gzip = T):
pdbs/4NP6.pdb exists. Skipping download


  |
  |                                                                      |   0%
  |
  |====                                                                  |   5%
  |
  |=======                                                               |  10%
  |
  |==========                                                            |  15%
  |
  |==============                                                        |  20%
  |
  |==================                                                    |  25%
  |
  |======================                                                |  30%
  |
  |========================                                              |  35%
  |
  |============================                                          |  40%
  |
  |================================                                      |  45%
  |
  |====================================                                  |  50%
  |
  |========================================                              |  55%
  |
  |============================================                          |  60%
  |
  |================================================                      |  65%
  |
  |====================================================                  |  70%
  |
  |========================================================              |  75%
```

```
  |
  |=================================================================|  80%
  |
  |=================================================================|  85%
  |
  |=================================================================|  90%
  |
  |=================================================================|  95%
  |
  |=================================================================| 100%
```

```
#path = declares where the data id downloaded, creates a new folder called "pdbs" for the pdb
```

Installing **BiocManager** from CRAN to install BioConducnctor. Using the install function in BiocManager to install Bioconducnctor: `BiocManager::install("msa")`

Aligning the PDB files to superimpose structures.

```
pdbs <- pdbaln(files, fit = TRUE, exefile="msa") # fit = true aligns positions, exefile = cal
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/8BQF_A.pdb
pdbs/split_chain/4X8M_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/9R6U_A.pdb
pdbs/split_chain/9R71_A.pdb
pdbs/split_chain/8Q2B_A.pdb
pdbs/split_chain/8RJ9_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/4X8H_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/8PVW_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/4NP6_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

Extracting sequences

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/8BQF_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbs/split_chain/6S36_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/9R6U_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbs/split_chain/9R71_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/8Q2B_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbs/split_chain/8RJ9_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbs/split_chain/6RZE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10   name: pdbs/split_chain/4X8H_A.pdb
pdb/seq: 11   name: pdbs/split_chain/3HPR_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 13   name: pdbs/split_chain/5EJE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 15   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 16   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 17   name: pdbs/split_chain/6HAM_A.pdb
```

```
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18   name: pdbs/split_chain/8PVW_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19   name: pdbs/split_chain/4K46_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20   name: pdbs/split_chain/4NP6_A.pdb
```

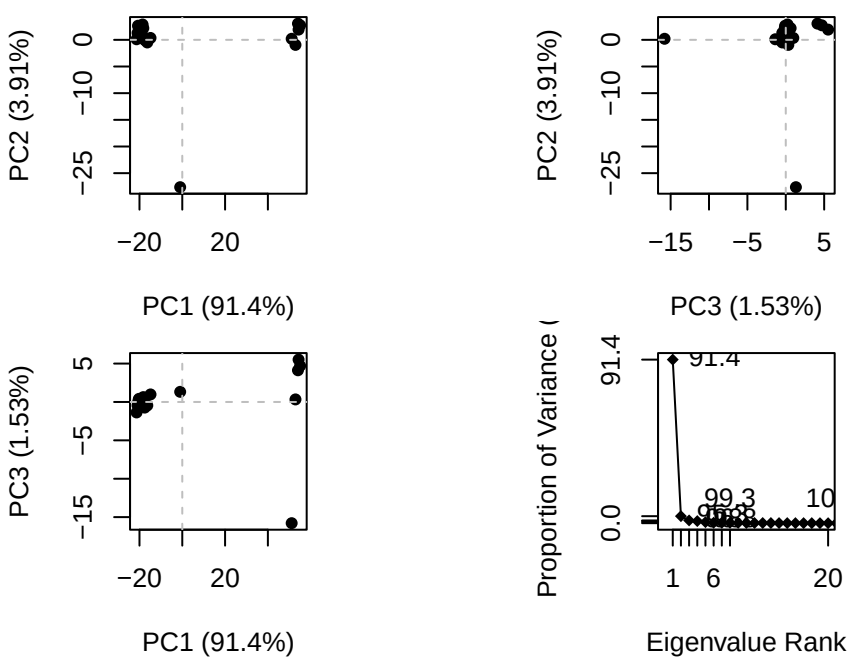
Viewing the strucutres with bio3view:

```
#view.pdbs(pdbs)
```

```
#view.pdbs(pdbs, colorScheme ="residue")
```

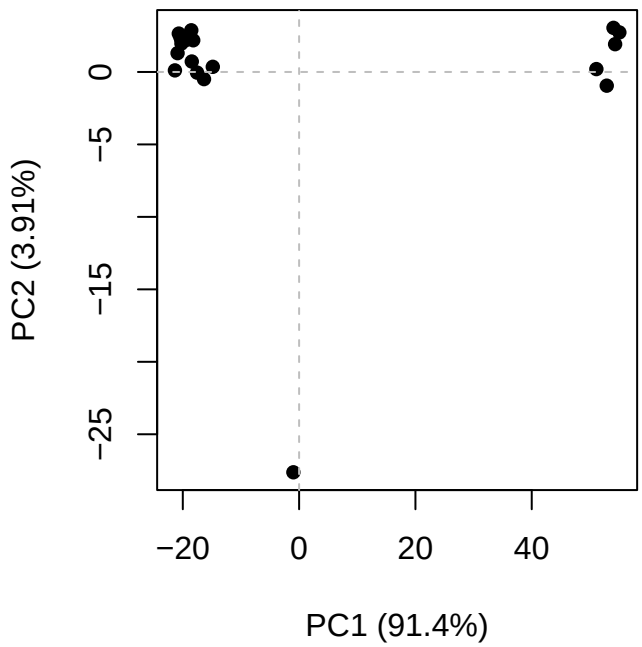
Running functions `rmsd()`, `rmsf()`, and `pca()` to display distance between atoms from each structure

```
pc.xray <- pca(pdbs)
plot(pc.xray)
```



Plotting PC1 vs PC2

```r
plot(pc.xray, 1:2)
```



Creating a movie to highlight the major motion of the dataset. Visualize in molstar

```r
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```